

# Operational Prioritization of Clinic Service Patterns Using Public Medicare Provider-Service Data

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## Abstract

Fast-growing ambulatory care organizations need reporting systems that do more than count visits after the fact. Operational leaders need to know where volume is concentrated, which services drive payment, which markets are changing, and which records should be reviewed first when data quality, cost, staffing, and clinical workflow questions compete for attention. This project evaluated whether public Medicare provider-service data can support a practical operational-priority score for clinic service patterns across Illinois and Indiana.

The analysis used the Centers for Medicare & Medicaid Services Medicare Physician and Other Practitioners by Provider and Service dataset for 2023 and 2024 [6, 7]. The selected service cohort included common outpatient evaluation and management codes, higher-acuity evaluation codes, electrocardiogram testing, and therapeutic procedure codes. A reproducible R workflow cleaned the data, created service-family and market summaries, exported SQL-ready tables, and ranked provider-service records using volume, estimated payment, service intensity, charge-to-allowed ratios, payment per service, and year-over-year growth. The final analytic file contained 237,405 provider-service records.

In 2024, the selected cohort included 51,101 providers, 861 city-state markets, 18,721,396 services, and \$1.18 billion in estimated Medicare payments. The highest operational-priority decile represented 10.0% of records but captured 40.0% of services and 42.1% of estimated payments. The findings support the hypothesis that transparent service, market, and payment features can identify a focused operational review cohort that captures a disproportionate share of activity.

## 1 Introduction

Clinic operations depend on a steady flow of small decisions. Leaders decide where staffing is tight, where patient access is changing, which service lines need clearer reporting, which locations need follow-up, and which operational metrics should be reviewed every week instead of once a quarter. The data behind those decisions often come from electronic health records, billing systems, finance files, marketing reports, inventory systems, and spreadsheets. Those systems are useful, but they are rarely clean by default. Even mature electronic health record environments require careful attention to completeness, consistency, documentation, and fitness for a specific analytic question [12, 8].

Public provider-service data cannot replace internal EMR, finance, or inventory data. It does, however, provide a useful way to practice the core analytic logic of clinic operations: define the service cohort, validate the fields, create stable summaries, identify high-volume and high-cost segments, and present results in a way that nontechnical leaders can use. CMS provider-service files include provider, location, HCPCS code, place-of-service, beneficiary, service, submitted charge, allowed amount, and payment fields [6, 4]. Those fields are not patient-level encounters, but they are close enough to show how operational reporting can be built from transactional healthcare data.

This project treats provider-service records as a public proxy for clinic service activity. The purpose is not to score clinical appropriateness or to infer whether a service should have occurred. The purpose is more practical: to create a transparent method for prioritizing records and markets for operational review. That kind of prioritization is consistent with broader health analytics work, where models and dashboards are most useful when they support decisions, monitoring, and follow-up rather than producing isolated statistics [2, 9, 10].

## **2 Research Question and Hypothesis**

The primary research question was: among selected clinic-relevant Medicare professional services in Illinois and Indiana, can transparent operational features identify a small review cohort that captures a disproportionate share of service volume and estimated payment?

The primary hypothesis was that an operational-priority score based on service volume, estimated payment, service intensity, charge-to-allowed ratios, payment per service, and year-over-year growth would concentrate more than 25% of 2024 service volume and estimated payment into the highest-priority decile.

A secondary descriptive question examined which service families, markets, and provider types contributed most to 2024 volume, estimated payment, and operational flags.

## **3 Data**

The project used the CMS Medicare Physician and Other Practitioners by Provider and Service public-use data for 2023 and 2024 [6]. The extract covered providers in Illinois and Indiana. The selected HCPCS cohort included new patient office visits (99203 and 99204), established patient office visits (99213, 99214, and 99215), follow-up or acuity-related visits (99232 and 99285), electrocardiogram testing (93000), and therapeutic procedures (97110). Physician fee schedule and HCPCS resources were used to interpret these as professional service records rather than patient-level encounter data [5, 4].

These codes were selected because they represent a compact service mix that can appear in ambulatory operations, referral follow-up, urgent care, diagnostic testing, therapy, and higher-acuity evaluation workflows. CMS data are aggregated by provider, service code, and place of service. They do not include patient identifiers, appointment dates, staffing levels, wait times, internal cost, inventory use, marketing source, payer mix beyond Medicare, or clinical notes. Those gaps matter because operational decisions inside a clinic network require internal data sources, but the public file is still useful for building and demonstrating a repeatable analytic workflow.

## **4 Methods**

### **4.1 Data Processing**

The R pipeline read the public CMS extract, standardized field types, joined HCPCS codes to service-family labels, and created derived operational fields. The main fields included rendering NPI, provider type, city, state, HCPCS code, place of service, total beneficiaries, total services, average submitted charge, average Medicare allowed amount, and average Medicare payment. Estimated totals were calculated by multiplying average dollar fields by total services:

estimated submitted charge = total services x average submitted charge  
estimated allowed amount = total services x average Medicare allowed amount  
estimated Medicare payment = total services x average Medicare payment

The pipeline also created services per beneficiary, payment per service, payment per beneficiary, and charge-to-allowed ratio. City and state were combined into a market field. Service codes were grouped into core visits, follow-up and acuity, and diagnostics and procedures. The project also exported a SQLite database and SQL views so that summaries could be rerun using stable tables. That structure reflects a basic operating principle of data quality work: reports are easier to trust when raw records, derived fields, and final summaries are separated and documented [1, 11, 3].

## 4.2 Operational-Priority Score

The operational-priority score was designed as a ranking tool for management attention. It is not a clinical risk score and it is not a compliance score. Each 2024 provider-service record was assigned percentile features for service volume, estimated Medicare payment, services per beneficiary, charge-to-allowed ratio, payment per service, and service growth from 2023 to 2024.

Volume, payment, and payment per service were ranked within year and state. Service intensity and charge-to-allowed ratios were ranked within year, state, HCPCS code, and place of service so that providers were compared with similar service records. Growth was ranked within year, state, and operations group. The score combined these features with higher weight on volume and payment because operational leaders usually need to know where the largest activity and financial exposure sit first.

30 percent volume percentile  
25 percent payment percentile  
15 percent service-intensity percentile  
10 percent charge-to-allowed percentile  
10 percent payment-per-service percentile  
10 percent service-growth percentile

Additional flags marked records with high volume, high intensity, and rapid growth. Records were divided into deciles within 2024. Decile 10 represented the records with the highest operational-priority scores.

## 4.3 Statistical Analysis

The analysis was descriptive and operational. The primary outcome was capture efficiency: the share of 2024 services and estimated payments contained in the highest operational-priority decile. The primary hypothesis was supported if the top decile captured more than 25% of services and estimated payments. Secondary analyses summarized service-family volume, payment, market concentration, provider-type concentration, growth, and flag counts.

No inferential model was used because the objective was not to estimate a causal effect. The goal was to build a defensible reporting workflow that could guide review and prioritization. In a production setting, the same approach could be extended with internal EMR encounter data, finance files, payer mix, staffing levels, inventory feeds, patient acquisition channels, and actual clinic locations.

## 5 Results

### 5.1 Analytic Cohort

The final analytic dataset included 237,405 provider-service records across 2023 and 2024. In 2024, the cohort represented 51,101 providers and 861 markets. Total 2024 service volume was 18,721,396, with \$1.18 billion in estimated Medicare payments. Estimated payment in the selected cohort changed by -2.0% from 2023 to 2024.

Table 1: 2024 service-family summary

| Operations group           | Service family                        | Records | Providers | Services   | Est. payment |
|----------------------------|---------------------------------------|---------|-----------|------------|--------------|
| Core visits                | Established patient office visit      | 70,722  | 35,767    | 10,592,323 | \$794.0M     |
| Core visits                | New patient office visit              | 23,807  | 17,496    | 1,102,143  | \$102.5M     |
| Diagnostics and procedures | ECG and diagnostic testing            | 3,203   | 3,203     | 381,874    | \$3.8M       |
| Diagnostics and procedures | Therapeutic procedure                 | 5,684   | 5,683     | 3,768,222  | \$64.0M      |
| Follow-up and acuity       | Emergency or high-acuity evaluation   | 4,088   | 4,088     | 633,701    | \$83.6M      |
| Follow-up and acuity       | Subsequent hospital or follow-up care | 12,000  | 11,957    | 2,243,133  | \$134.2M     |

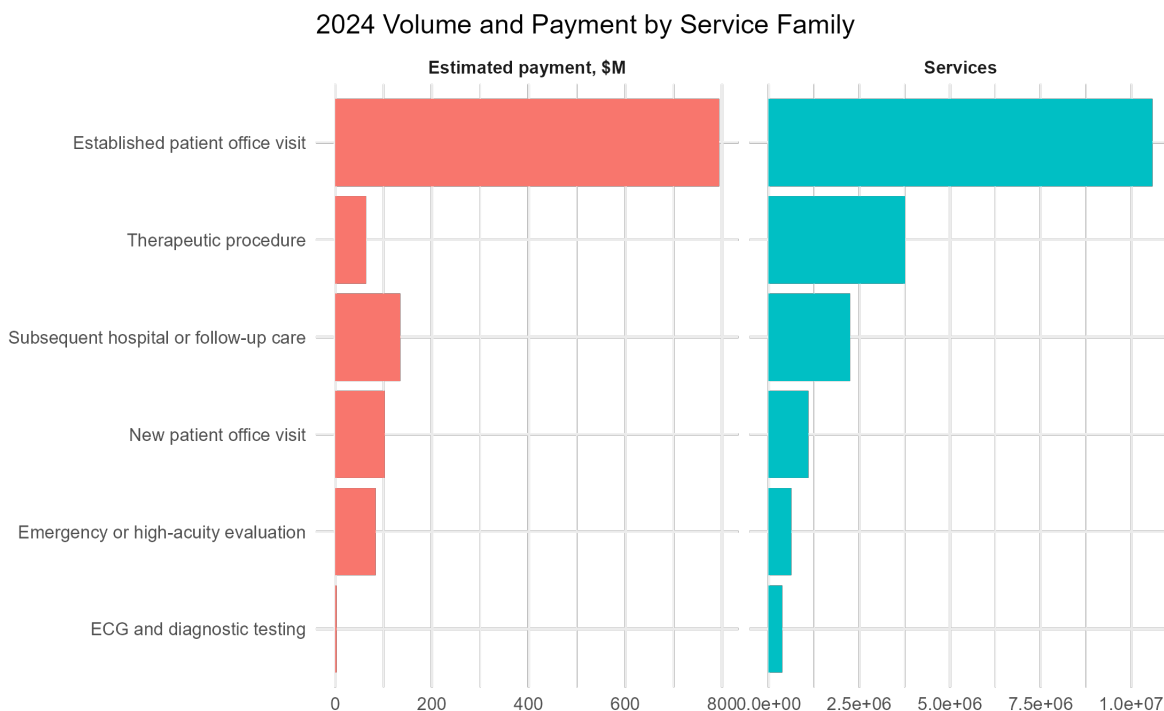


Figure 1: Service volume and estimated Medicare payment by service family in 2024.

## 5.2 Operational-Priority Capture

The highest operational-priority decile represented 10.0% of 2024 records. It captured 40.0% of services and 42.1% of estimated Medicare payments. This exceeded the 25% threshold specified in the primary hypothesis. It also means that a focused review of one-tenth of provider-service records would cover roughly four-tenths of the selected service activity.

Table 2: 2024 operational-priority decile summary

| Decile | Records | Services  | Est. payment | Service share | Payment share |
|--------|---------|-----------|--------------|---------------|---------------|
| 1      | 11,950  | 190,659   | \$10.7M      | 1.0%          | 0.9%          |
| 2      | 11,950  | 261,720   | \$17.4M      | 1.4%          | 1.5%          |
| 3      | 11,951  | 373,989   | \$24.7M      | 2.0%          | 2.1%          |
| 4      | 11,950  | 528,354   | \$34.8M      | 2.8%          | 2.9%          |
| 5      | 11,951  | 751,494   | \$48.7M      | 4.0%          | 4.1%          |
| 6      | 11,950  | 1,090,295 | \$69.2M      | 5.8%          | 5.9%          |
| 7      | 11,950  | 1,587,433 | \$99.1M      | 8.5%          | 8.4%          |
| 8      | 11,951  | 2,431,814 | \$145.2M     | 13.0%         | 12.3%         |
| 9      | 11,950  | 4,011,073 | \$234.6M     | 21.4%         | 19.8%         |
| 10     | 11,951  | 7,494,565 | \$497.9M     | 40.0%         | 42.1%         |

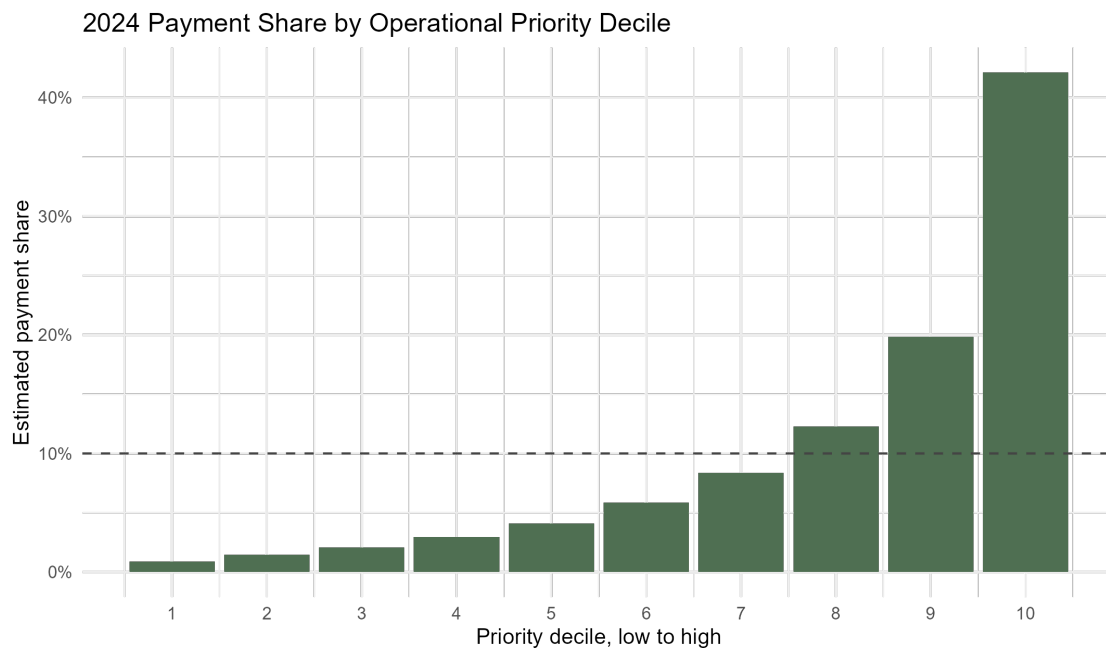


Figure 2: Estimated payment share by operational-priority decile in 2024.

## 5.3 Market Patterns

The largest 2024 market in this extract was Chicago, Illinois, with 1,992,025 selected services and \$142.2 million in estimated payment. Chicago was expected to be the largest market because the analysis included all selected Illinois and Indiana provider-service records, not only urgent-care centers. The market table is

still useful because it shows how an operational report can rank local service activity, identify concentration, and support follow-up questions about access, staffing, and service mix.

Table 3: Top 2024 markets by selected service volume

| Market           | Providers | Service families | Services  | Est. payment |
|------------------|-----------|------------------|-----------|--------------|
| Chicago, IL      | 7,961     | 6                | 1,992,025 | \$142.2M     |
| Indianapolis, IN | 3,682     | 6                | 984,193   | \$63.6M      |
| Fort Wayne, IN   | 1,557     | 6                | 383,824   | \$24.5M      |
| Evansville, IN   | 871       | 6                | 357,457   | \$21.8M      |
| Springfield, IL  | 1,027     | 6                | 322,346   | \$22.1M      |
| Peoria, IL       | 948       | 6                | 316,994   | \$21.0M      |
| Naperville, IL   | 763       | 6                | 282,776   | \$19.8M      |
| Rockford, IL     | 762       | 6                | 236,088   | \$13.7M      |

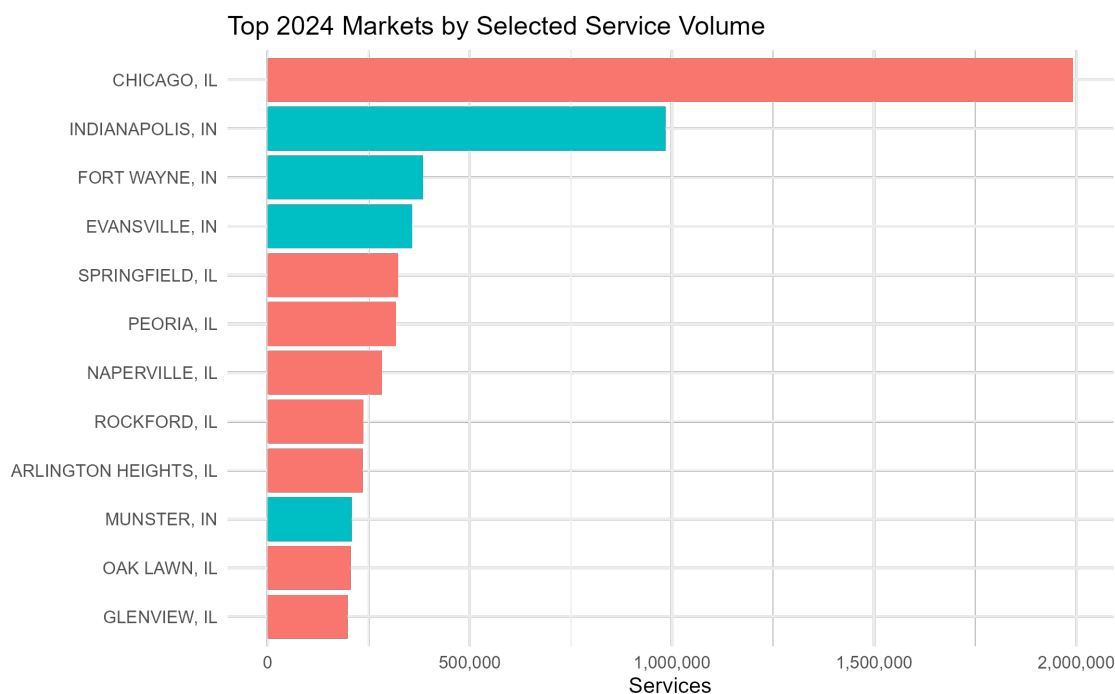


Figure 3: Top 2024 markets by selected service volume.

## 5.4 Service Growth and Provider Mix

Growth flags identified provider-service records where service volume and estimated payment both increased by more than 30% from 2023 to 2024. These flags do not imply that growth is good or bad. They identify records where operations leaders may want context. Growth could reflect changes in patient demand, referral patterns, coding mix, clinic access, provider staffing, or documentation behavior.

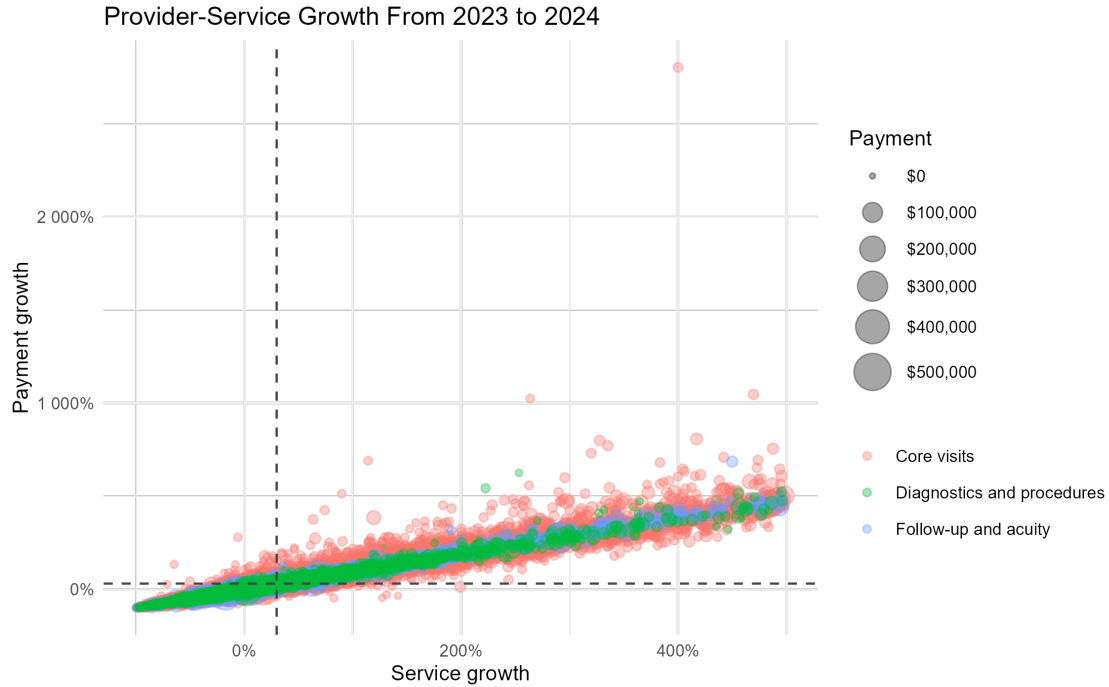


Figure 4: Provider-service volume and payment growth from 2023 to 2024.

Provider-type summaries help separate market-level volume from workforce or specialty mix. In the selected 2024 cohort, the largest provider types included physical therapists in private practice, nurse practitioners, internal medicine, family practice, cardiology, physician assistants, emergency medicine, and orthopedic surgery.

Table 4: Provider types with the largest 2024 service volume

| Provider type                          | Providers | Services  | Est. payment | Growth flags |
|----------------------------------------|-----------|-----------|--------------|--------------|
| Physical Therapist in Private Practice | 4,743     | 3,274,513 | \$55.5M      | 875          |
| Nurse Practitioner                     | 9,886     | 2,128,455 | \$132.2M     | 3,957        |
| Internal Medicine                      | 4,531     | 1,965,776 | \$144.0M     | 1,580        |
| Family Practice                        | 4,577     | 1,640,567 | \$120.4M     | 1,750        |
| Cardiology                             | 992       | 833,351   | \$56.9M      | 666          |
| Physician Assistant                    | 3,914     | 780,005   | \$49.9M      | 1,635        |
| Emergency Medicine                     | 3,072     | 609,480   | \$78.3M      | 571          |
| Orthopedic Surgery                     | 1,148     | 524,158   | \$38.5M      | 691          |

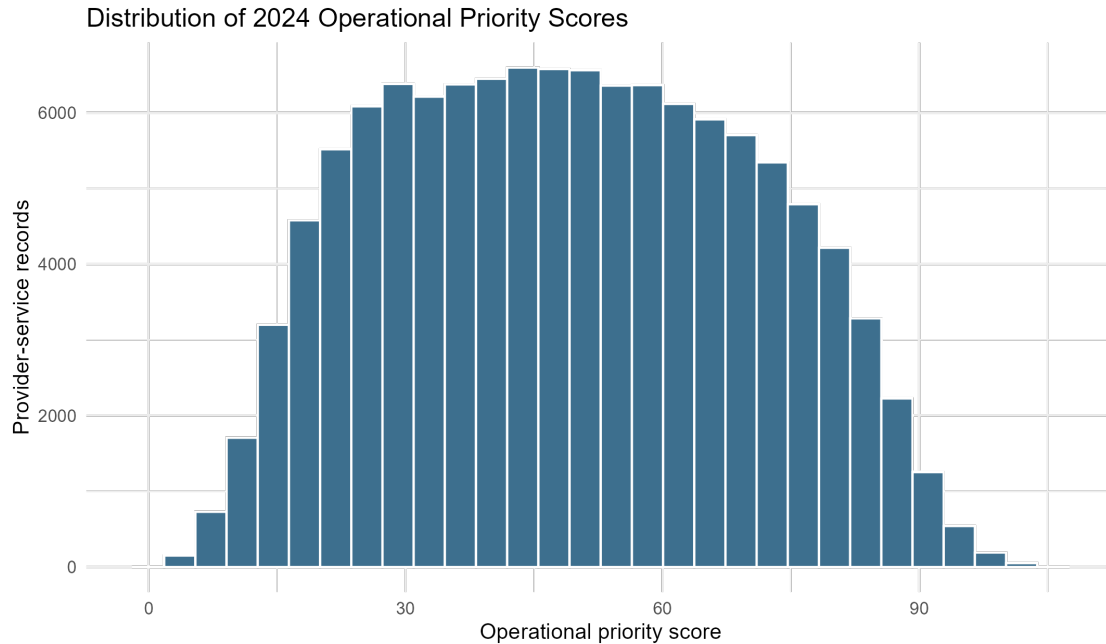


Figure 5: Distribution of 2024 operational-priority scores.

## 6 Discussion

The primary hypothesis was supported. A transparent operational-priority score concentrated a substantial share of service volume and estimated payment into a small review cohort. The top decile captured 40.0% of 2024 services and 42.1% of estimated payments while representing only 10.0% of records. That is the main operational result: a simple score can turn a large provider-service file into a focused list for leaders or analysts to review.

The practical value is not only the score. The workflow also produces reusable tables by service family, market, provider type, code, and priority decile. Those outputs resemble the building blocks of an internal clinic reporting system. A leadership team could use similar tables to ask which markets are growing, which services drive volume, which service families need better definitions, which records need data-quality review, and where follow-up with clinical or finance teams is warranted.

The project also shows why healthcare operations analytics should avoid black-box reporting when the audience is cross-functional. Clinical, finance, and operations leaders need to know why something was flagged. A record can be surfaced because it has high service volume, high estimated payment, high service intensity, high payment per service, rapid growth, or some combination of those features. That interpretability matters in healthcare analytics because operational data are shaped by workflow, coding, documentation, access, and policy [12, 8].

The selected CMS data are not internal EMR data, but the same method would transfer naturally to an internal environment. EMR appointment tables could supply no-show rates, visit status, provider schedules, and cycle-time metrics. Finance files could supply payer mix, reimbursement, and cost. Marketing systems could add acquisition channels. Inventory systems could add supply and stockout metrics. The main analytic pattern would remain the same: clean the source tables, define operational metrics, write stable SQL summaries, rank priorities, and communicate the results plainly.



## 7 Limitations

This project used public Medicare provider-service aggregates, not patient-level EMR records. It does not include appointment timestamps, patient wait time, rooming time, visit status, payer mix beyond Medicare, internal cost, inventory units, marketing source, or staffing schedules. It also does not identify urgent-care centers directly. Because of those limits, the findings should be interpreted as a demonstration of operational analytics logic rather than a complete clinic-performance assessment.

The service cohort was intentionally compact. It included common evaluation and management services, follow-up or acuity-related services, ECG, and therapy. Other urgent or primary care services, such as lab testing, imaging, vaccinations, injections, occupational medicine, and procedure-specific supplies, would require a broader code set or internal operational data. The cohort also covered Illinois and Indiana only, which makes the geographic scope relevant for Midwest service patterns but not national operations.

Finally, estimated Medicare payments are not the same as clinic revenue, contribution margin, or internal cost. Medicare payment can be useful for public benchmarking, but internal finance files would be needed for actual cost optimization. The operational-priority score should be treated as a triage tool for review, not as a final judgment about performance.

## 8 Conclusion

Public provider-service data can support a useful prototype for clinic operations analytics. In this Illinois and Indiana service cohort, a transparent operational-priority score identified a top decile that captured 40.0% of services and 42.1% of estimated payments. The project demonstrates how healthcare data can be cleaned, structured, summarized with SQL-ready views, and translated into market, service-line, and review-priority outputs that an operations team could inspect.

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